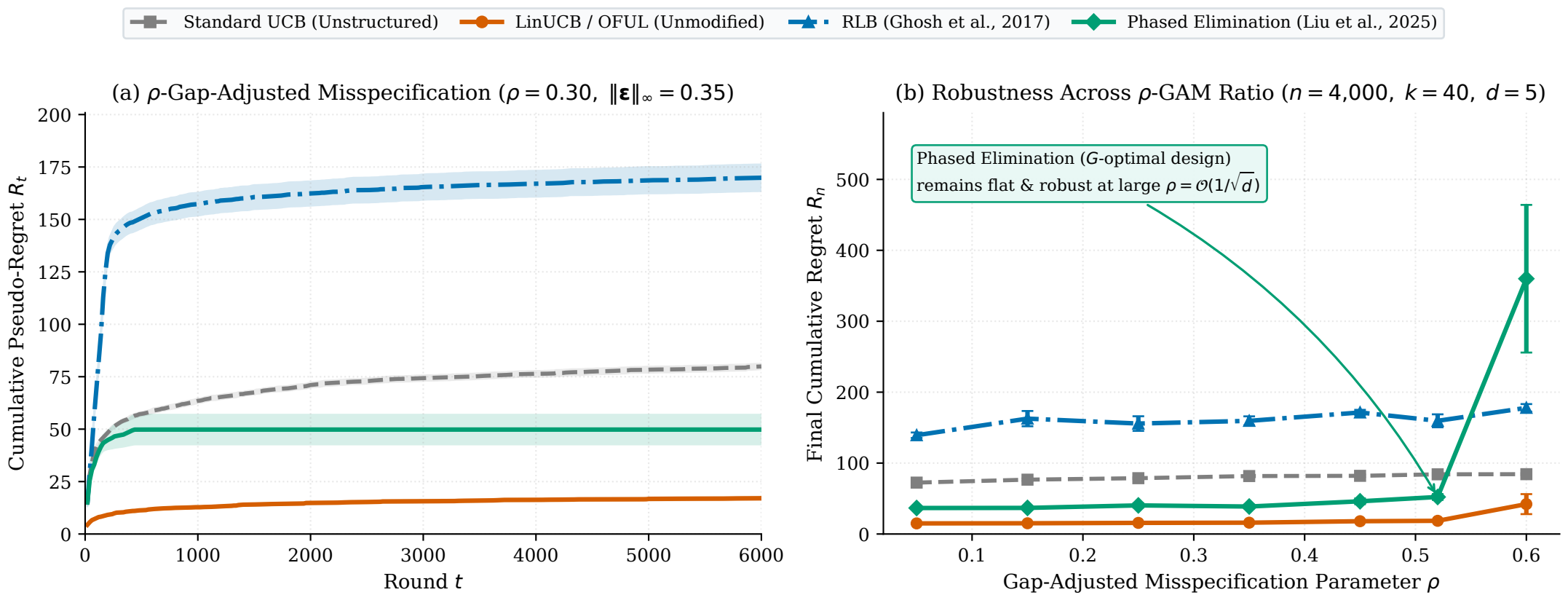

Standard UCB (Unstructured)
LinUCB / OFUL (Unmodified)
RLB (Ghosh et al., 2017)
Phased Elimination (Liu et al., 2025)
(a) ρ-Gap-Adjusted Misspecification (ρ = 0.30, ‖ε‖∞ = 0.35)
Cumulative Pseudo-Regret R_t
Round t
0
25
50
75
100
125
150
175
200
0
1000
2000
3000
4000
5000
6000
(b) Robustness Across ρ-GAM Ratio (n = 4,000, k = 40, d = 5)
Final Cumulative Regret R_n
Gap-Adjusted Misspecification Parameter ρ
Phased Elimination (G-optimal design)
remains flat & robust at large ρ = 𝒪(1/√d)
0
100
200
300
400
500
0.1
0.2
0.3
0.4
0.5
0.6